

Shared evidence | Checkpoint cases and measurements

Health and plants: evidence checkpoint

Name _____ Date _____

Use these original cases and constructed sample data. No personal health information is needed.

Warm-up only (W1-W3): P has a rash caused by a transmissible virus; Q has a rash from a non-infectious skin reaction. W1: does the same symptom establish the same cause? W2: an antibody is a protein; a memory lymphocyte is a cell. Explain the difference. W3: separate practice counts are 7, 8 and 9 bubbles, each over 2 minutes. Find the mean and then the rate per minute. Use the checkpoint cases below only for the independent task.

Case A	Case B
A fictional respiratory illness is caused by a virus that can pass between hosts. Coughing may occur.	An inherited airway condition is not caused by a transmissible infectious agent. Coughing may occur.

Light sensor (lux)	Three bubble counts	Minutes per count
200	12,15,18	3 each
800	30,33,36	3 each

Constructed pondweed investigation: same shoot, solution and temperature at both light levels. Each count lasts 3 minutes. Bubble volumes may differ and some oxygen dissolves.

Record the route used and any read-aloud, scribing or additional support.

Supported route | Four scientific checks

Original 16-mark checkpoint. Use the shared cases and constructed data. Each numbered question is worth 4 marks. Record the route/support; this is not an official exam paper. Access word bank: cause; feature; protein; cell; reactant; tissue; time; proxy.

S1. Classify cases A and B as communicable or non-communicable [2]. Give the causal reason for A [1]. Explain why coughing alone cannot decide [1].

S2. Define antigen [1] and antibody [1]. Identify memory lymphocytes as cells [1] and explain their role in a later response to the same antigen [1].

S3. Name both material reactants of photosynthesis [1] and its sugar product [1]. Name the tissue carrying water/mineral ions [1] and the tissue carrying sucrose [1].

S4. For 800 lux, calculate the mean count [1] and bubbles per minute [1]. Name one control for comparing the two light levels [1] and one reason count is not exact oxygen volume [1].

Standard route | Explain and calculate

Original 16-mark checkpoint. Use the shared cases and constructed data. Each numbered question is worth 4 marks. Record the route/support; this is not an official exam paper.

M1. Classify cases A and B as communicable or non-communicable [2]. Give the causal reason for A [1]. Explain why coughing alone cannot decide [1].

M2. Define antigen [1] and antibody [1]. Identify memory lymphocytes as cells [1] and explain their role in a later response to the same antigen [1].

M3. Name both material reactants of photosynthesis [1] and its sugar product [1]. Name the tissue carrying water/mineral ions [1] and the tissue carrying sucrose [1].

M4. For 800 lux, calculate the mean count [1] and bubbles per minute [1]. Name one control for comparing the two light levels [1] and one reason count is not exact oxygen volume [1].

Challenge route | Precise explanations

Original 16-mark checkpoint. Use the shared cases and constructed data. Each numbered question is worth 4 marks. Record the route/support; this is not an official exam paper. Answer without the access word bank; justify each causal or measurement distinction.

T1. Classify cases A and B as communicable or non-communicable [2]. Give the causal reason for A [1]. Explain why coughing alone cannot decide [1].

T2. Define antigen [1] and antibody [1]. Identify memory lymphocytes as cells [1] and explain their role in a later response to the same antigen [1].

T3. Name both material reactants of photosynthesis [1] and its sugar product [1]. Name the tissue carrying water/mineral ions [1] and the tissue carrying sucrose [1].

T4. For 800 lux, calculate the mean count [1] and bubbles per minute [1]. Name one control for comparing the two light levels [1] and one reason count is not exact oxygen volume [1].

Feedback and next step | Keep first and revised answers

Complete this after the independent checkpoint. Corrections are not added to the first-attempt score.

L1. Copy one short part of your first answer. Improve it and explain why the change is scientifically stronger.

E1. State one distinction you can now explain accurately.

E2. Choose one specific next teaching target and say what would help.

My next useful focus

- A. Classify from cause and transmission.
- B. Separate antibody proteins and memory cells.
- C. Match plant materials with tissues.
- D. Calculate a rate and state its proxy limit.